

YUBO LI

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EDUCATION

Carnegie Mellon University - <i>Ph.D. in Information Systems</i>	2022 – Present
Carnegie Mellon University - <i>M.S. in Information Systems & Management</i>	2020 – 2021
University of California, San Diego - <i>B.S. in Applied Mathematics Minor in Business</i>	2015 – 2019
<i>Leadership: Founder of the UCSD Triton Actuarial Society GPA: 3.8/4.0</i>	

PUBLICATIONS

- [Confidence as Control: A Survey of Confidence Utilization in Large Language Models](#). TMLR
- [Consistency of Large Reasoning Models Under Multi-Turn Attacks](#). EMNLP 2026
- [Same Question, Different Source, Different Answer: Auditing Source-Dependence in Medical Multi-Source RAG](#). EMNLP 2026
- [The Chain Holds, the Answer Folds: Trace-Answer Dissociation in Reasoning Models Under Adversarial Pressure](#). EMNLP 2026
- LLMs Know the Constraint But Do Not Use It: Activation Bottlenecks in Pragmatic Constraint Reasoning. EMNLP 2026
- [The Model Says Walk: How Surface Heuristics Override Implicit Constraints in LLM Reasoning](#). COLM 2026
- [Skills-Bench : Benchmarking the Efficacy of Agent Skills Across Diverse Tasks](#), NeurIPS 2026
- [PANDO: Efficient Multimodal AI Agents via Online Skill Distillation](#). NeurIPS 2026 (Under review)
- [Beyond Single-Turn: A Survey on LLM Multi-Turn Interactions](#). TMLR 2026.
- [Time-To-Inconsistency: A Survival Analysis of Large Language Model Robustness to Adversarial Attacks](#). ICLR 2026.
- [Firm or Fickle? Evaluating LLM Consistency Capability in Multi-Turn Interactions](#). ACL 2025.
- [No Black Box Anymore: Demystifying Clinical Predictive Modeling with Temporal-Feature Cross Attention Mechanism](#). American Medical Informatics Association 2025 Annual Symposium (3rd Place in Paper Competition).
- [Enhancing End Stage Renal Disease Outcome Prediction: A Multi-Sourced Data-Driven Approach](#). Journal of the American Medical Informatics Association, 2025.
- [Co-Alignment: Rethinking Alignment as Bidirectional Human-AI Cognitive Adaptation](#). Agents4Science 2025.
- [Towards Interpretable End-Stage Renal Disease \(ESRD\) Prediction: Utilizing Administrative Claims Data with Explainable AI Techniques](#). American Medical Informatics Association 2024 Annual Symposium.

FELLOWSHIPS

- 2025 Presidential Fellowship, Tata Consultancy Services (TCS)
- [2024 Fellowship in Generative AI in Healthcare](#), Center for Machine Learning and Health, Carnegie Mellon University
- [2023 Fellowship in Digital Health Innovation](#), Center for Machine Learning and Health, Carnegie Mellon University

RESEARCH EXPERIENCE

- **Efficient AI Agents | Agent Skills** **Carnegie Mellon University**
Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2025.01 - Present
 - **AI Agents**: Built PANDO, a single-rollout online skill-distillation framework that equips multimodal web agents with a structured Skill Library via progress reflection, confidence-based skill demotion, hierarchical routing, visual compression, and cache-aware prompting; reached 58.3% success on the full 910-task VisualWebArena while using 58-61% fewer tokens than prior agents (SGV, WALT).
 - **Agent Skills Benchmarking**: Co-developed SkillsBench, a paired no-Skills versus curated-Skills benchmark spanning 87 tasks across 8 domains with deterministic verifiers, showing curated skills lift average pass rate from 33.9% to 50.5% (+16.6 pp) and that focused, three-module skills outperform exhaustive bundles; also surveyed dynamic, self-evolving agent skills.
 - **Efficient Inference**: Designed CONF-KV, a confidence-aware KV-cache manager with mixed FP16/INT8 storage that sets per-step cache budgets from model uncertainty, achieving 91.4% Needle-in-a-Haystack retrieval at 32K tokens (versus 53.8% for sliding windows) at 2.8x lower peak memory.
- **Q&A Systems for Trustworthy Organ Transplantation Guidance** **Carnegie Mellon University**
Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2025.01 - Present
 - **Q&A Systems**: Developed Q&A systems for organ transplantation patient care by integrating retrieval-augmented generation (RAG) with adaptive frameworks that synthesize patient-specific medical history, clinical guidelines, and dynamic risk factors to deliver personalized guidance and support.
 - **RAG/Retrieval**: Engineered a hybrid retrieval pipeline combining both sparse and dense retrievers (DPR, ANCE) to accurately extract and verify medical content from over 100 transplant-specific handbooks.
 - **LLM Reasoning**: Developed an incentivized reasoning mechanism that align LLM behaviors by rewarding multi-step chain-of-question sequences, enabling the system to reason freely and proactively request critical diagnostic information.
 - **Source Auditing**: Built a framework to audit source-dependence in multi-source medical RAG, releasing TransplantQA and a hierarchical retrieve-and-audit strategy (HERO-QA) scored by a validated five-label inter-source taxonomy; across 102 handbooks from 23 transplant centers, quantified that 20.8% of comparable answers diverge clinically while 96.2% of question-source pairs reveal coverage gaps.

- LLM Alignment & Multi-Turn Consistency

Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor

2024.05 - Present

Distributed Training:

Architected and optimized distributed training infrastructure for LLMs (Mistral, LLaMA, Deepseek) across 4 nodes with 16 NVIDIA GH200 GPUs, implementing tensor parallelism, pipeline parallelism, and ZeRO optimizer techniques to efficiently scale model training and reduce computational overhead.

Fine-tuning:

Performed fine-tuning of state-of-the-art LLMs both locally (on-premises clusters) and via OpenAI's official GPT-4o API, successfully improving model consistency, confidence-aware reasoning, and achieving state-of-the-art performance.

Evaluation:

Proposed a metric to quantitatively assess multi-turn response stability, prioritizing early interaction accuracy and swift recovery from initial errors. Released a comprehensive benchmark to test the model's ability to sustain coherent and contextually accurate responses during dynamic, multi-turn conversations.

Faithfulness & Constraints:

Documented unfaithful capitulation, where a correct chain-of-thought persists while the emitted answer flips wrong under pressure, and introduced the Heuristic Override Benchmark showing salient surface cues override unstated feasibility constraints; released the supporting benchmarks, trajectories, and judge labels.

WORKING EXPERIENCE

- Netflix – ML Research Scientist Intern

2026.06 – 2026.08
- Personalized GenRec:

Developed personalized generative search models that jointly encode free-text queries and user interaction history to produce ranked semantic-ID recommendation lists, improving MRR@50 from 0.22 to 0.67.
- Continued pretraining & post-training:

Built a two-stage adaptation pipeline—continued pretraining on domain corpora followed by multi-task post-training—to equip the LLM with query-only retrieval, semantic-ID retrieval, instruction following, and reasoning, using a loss-weighting scheme to balance the competing objectives.
- Causal Evaluation & Robustness:

Proposed counterfactual evaluation protocols to disentangle whether models learn to use user history causally versus memorizing history patterns, and stress-tested robustness under noisy-history interventions.
- Amazon – Applied Scientist Intern

2025.06 – 2025.08
- Generative RecSys:

Developed User-LLM pipeline for personalized recommendation by pre-training custom user encoder on 172k customer interaction sequences, then fine-tuning through progressive alignment strategies (adapter-only, adapter + LLM, full fine-tuning) and pooling mechanisms.
- Achieved superior predictive performance (60.1% Hit Ratio@3, 67.5% NDCG@3) compared to direct prompting baselines while delivering **93.4% token reduction** and **64.8% cost savings**.
- RAG/Search:

Architected production-ready RAG service infrastructure using FAISS HNSW indexing for 60k user embeddings to enhance recommendation contest at scale.
- Distributed Training:

Engineered distributed training across multi-GPUs using FSDP with mixed-precision BF16.

SELECTED ACADEMIC ACTIVITIES

- Invited Talk & Guest Lecture

POMS 2026 | Invited Tutorial on LLM/AI Agent Measurement Science & Evaluations

GenAI in Healthcare | 94-706 Healthcare Information Systems| Spring 26

Coding Agents | 95-891 Introduction to AI | Spring 26

AI Agents | 95-891 Introduction to AI | Fall 25/Spring 26

Deep Learning for NLP | 95-891 Introduction to AI | Spring 25
- Teaching Assistant

11-711 Advanced NLP with Prof. Graham Neubig.

10-680/10-681 Mathematical & Computational Foundation for Machine Learning with Prof. Henry Chai.

95-891 Introduction to Artificial Intelligence with Prof. David Steier.

95-888 Data Focused Python with Prof. Michael Simko.

95-865 Unstructured Data Analysis with Prof. George Chen.

95-760 Decision Making under Uncertainty with Prof. David Choi.
- Courses

10-714 Deep Learning Systems

10-715 Advanced Introduction to Machine Learning

10-716 Advanced Machine Learning: Theory and Methods

10-703 Deep Reinforcement Learning

10-725 Convex Optimization

10-805 Machine Learning with Big Data

15-513 Introduction to Computer System

15-688 Practical Data Science

11-868 LLM Systems

11-711 Advanced NLP

11-785 Introduction to Deep Learning

11-777 Multi-Model Machine Learning

36-707 Regression Analysis

36-731/36-732 Causal Inference